1. **Lightweight Feistel-Based Smart Meter Encryption Suite**

- **Cryptographic Concepts:** Feistel Ciphers, 8-bit/64-bit Data Encryption Standards (DES), Symmetric ciphers.
- **Smart Grid Application:** Design a custom, resource-constrained 8-bit Feistel cipher simulation for low-power microcontroller-based residential smart meters. The project evaluates power consumption, execution time, and ciphertext diffusion when encrypting real-time kilowatt-hour usage reports sent to the utility provider.

2. **ECC-Driven Mutual Authentication for Electric Vehicle-to-Grid (V2G) Systems**

- **Cryptographic Concepts:** Elliptic curves over finite fields, Elliptic Curve Discrete Logarithm Problem (ECDLP), Elliptic Curve Cryptography (ECC).
- **Smart Grid Application:** Build a secure session establishment protocol between an Electric Vehicle (EV) and a charging station. The system uses finite field scalar multiplication and ECDLP hardness to ensure that malicious actors cannot spoof vehicle identities during high-power billing sessions.

3. **Post-Quantum GGH Lattice-Based Substation Command Integrity Framework**

- **Cryptographic Concepts:** GGH lattice-based digital signatures, Modular polynomial arithmetic, Information Theory.
- **Smart Grid Application:** Implement a lattice-based signature verification engine to protect supervisory control and data acquisition (SCADA) commands sent from control rooms to electrical substation circuit breakers, securing critical infrastructure against future quantum computing attacks.

4. **Scalable KDC Key Establishment Using Extended Euclidean RSA Management**

- **Cryptographic Concepts:** Extended Euclidean algorithm, RSA algorithm, Key Distribution Center (KDC), Modular arithmetic.
- **Smart Grid Application:** Construct a hierarchical Key Distribution Center (KDC) for an Advanced Metering Infrastructure (AMI) network. The system uses the Extended Euclidean algorithm to compute modular multiplicative inverses for automated, dynamic RSA public-private key pair generation across thousands of smart meters.

5. **Privacy-Preserving Aggregate MACs and Hash-Based Metering Verification**

- **Cryptographic Concepts:** Message Authentication Codes (MACs), Hash functions (SHA/MD5), Information Theory.

- **Smart Grid Application:** Develop an intermediate grid-aggregator module that validates aggregated power consumption data streams from entire neighborhoods. Using nested MACs and cryptographic hash chains, the aggregator verifies data integrity without inspecting individual household consumption profiles, preserving consumer privacy.

6. **Entropy and Information Theory Analysis of Grid PRNG Challenge-Response Protocols**

- **Cryptographic Concepts:** Random Number Generators (RNG), Pseudorandom Number Generators (PRNG), Information Theory.
- **Smart Grid Application:** Build a security testing framework that measures the entropy and unpredictability of PRNG-generated challenge tokens used in smart meter authentication handshakes, identifying vulnerabilities to replay attacks under simulated network jitter.

7. **Decentralized Microgrid Energy Trading Engine Using El Gamal and RSA Signatures**

- **Cryptographic Concepts:** RSA digital signatures, El Gamal digital signatures, Primitive roots, Euler's theorem.
- **Smart Grid Application:** Create a peer-to-peer (P2P) local energy trading simulator where prosumers sign smart contracts and localized power-transfer vouchers. The system contrasts the signature sizes, verification speeds, and mathematical overhead of RSA versus El Gamal in constrained edge nodes.

8. **Legacy Cipher Migration Framework for Industrial Control Systems (ICS)**

- **Cryptographic Concepts:** Monoalphabetic ciphers (Caesar, Affine, Additive), Polyalphabetic ciphers (Playfair, Vigenère), Advanced Encryption Standards (AES).
- **Smart Grid Application:** Construct a protocol-translation gateway that wraps legacy, unencrypted or classically encrypted telemetry data (simulating historical substitution/transposition ciphers) into modern AES-encrypted payloads, studying the transition vulnerabilities of upgrading legacy grid hardware.

9. **Lenstra's ECM Security Auditing Tool for Smart Grid Public Key Infrastructures (PKI)**

- **Cryptographic Concepts:** Prime numbers, Unique factorization, Lenstra's elliptic curve factorization algorithm (ECM), PKI.

- **Smart Grid Application:** Develop an automated vulnerability scanner utilizing Lenstra's ECM algorithm to audit the cryptographic strength of public keys deployed across field devices, flagging weak moduli or improper prime generation in legacy utility deployments.

10. **Modular Polynomial Diffie-Hellman Group Key Agreement for Neighborhood Area Networks (NAN)**

- **Cryptographic Concepts:** Diffie-Hellman Key Exchange, Discrete logarithms, Powers and primitive roots in finite fields, Modular polynomial arithmetic.
- **Smart Grid Application:** Implement a scalable group key establishment protocol over finite fields to secure broadcast and multicast communication channels among smart meters clustered within a shared Neighborhood Area Network (NAN).